Candidate Seat No:_________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Bachelor of Science in Information Technology (B.Sc. IT)
Semester-V University Examination November-2019
CA606 Advanced Data Science

Date: 22.11.19, Friday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **The figure to the right indicates marks.**
3. **Make a suitable assumption when necessary.**
4. **Simple Calculator is permitted.**

SECTION-1

Q.1	**Do as directed.** **1.** Define : Big Data**.** **2**. List out any four sources of semi structured data. **3**. Give any two names of personal assistant applications basded on NLP. **4**. What do you mean by machine generated data? **5**. _______ variables are variables that have two or more categories, but which do not have an intrinsic order. **6**. ____________ means to understand the data before further processing. **7**. What are the ways of univariate analysis for continuous variable? **8**. Which tests produce statistical significance to explore the relationship between categorical and continuos variables? **9**. The measurable characteristic of the ________ is called a statistic. **10**. _________ sampling technique is used in the situations where the population is completely rare or unknown. **11**. What do you mean by stratified sampling?	**[11]**
Q.2 [A]	Explain Four V's characteristics of Big Data.	**[04]**
[B]	Explain Data Preparation step of data science with all sub steps.	**[05]**
[C]	Give difference between population and sample.	**[03]**

OR

Q.2 [A]	Discuss the usage and benefits of data science and big data in education and banking domains.	**[04]**
[B]	What is the significance of Bivariate analysis? Discuss the possible combination of Bivariate analysis for continuous and categorical data.	**[05]**
[C]	Explain NLP in detail with its use cases.	**[03]**
Q.3	**Answer the following questions in detail** (**Any Four**). **1.** Explain different kinds of unstructured data with proper example. **2.** How to set the research goals of any data science problem? Discuss it for the problem "Distinguishing between noise and signal on millions of NASA pictures or videos is time consuming and complex task". **3**. Give difference between probability and non-probability sampling with appropriate parameters.	**[12]**

4. What are the characteristics of semi structured data? Give advantages of such data.
5. Discuss any three techniques of retrieving the data for any data science problem.
6. Which statistical model is suitable to determine the population growth of India? Discuss it in details.

SECTION-2

Q.4 **Do as directed.** **[11]**

1. What do you mean by reinforcement learning?
2. __________is a classification algorithm that is used to predict binary classification.
3. What is the significance of null deviance in any prediction model?
4. The confusion matrix of one problem is as under :

45 (TP)	4(FN)
5(FP)	48(TN)

Find the Recall of this situation.
5. Give the equation of posterior probability.
6. _________based filtering algorithm recommends products which are similar to the ones that a user has liked in the past.
7. __________________ is information that is provided intentionally, i.e. input from the users such as movie ratings in recommender system.
8. Google's page rank uses _____________________________centrality.
9. _________deals with the separation of the higher-level semantics of the system from the lower-level implementation issues.
10. List out any four complicating factors of the distributed data processing.
11.________ is an open source distributed stream processing & messaging Platform.

Q.5 [A] Determine three similar clusters based on the k-means clustering algorithm of following situation : **[05]**

Year	3	7	12	2	1	6	5	10	15	3	4	8	7	3	8
Return(%)	1	8	15	2	2	7	6	11	15	2	3	9	7	3	7

Take first three rows as a random clusters.

[B] Use cosine similarity to decide whether should we give the recommendtation to the user1 for movie-5. **[05]**

	Movie-1	Movie-2	Movie-3	Movie-4	Movie-5
User-1	5	4	4	3	?
User-2	1	2	1	1	2
User-3	4	4	4	4	4
User-4	5	3	4	4	4
User-5	1	2	1	2	1

[C] Explain how SVM works? **[02]**

OR

Q.5 [A] What is the objective of the Naïve Bayes algorithm? What are the applications of it? Give the equation of it. Write an R code to classify the fruits based on Long,Sweet and Yellow parameters (Make necessary assumptions while writing a code). **[05]**

[B] What is the objective of recommendation system? Explain the steps of working of recommendation systems. **[05]**

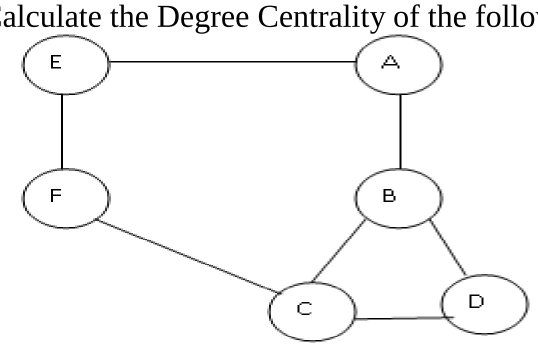
[C] Give brief about following machine learning terminologies: **[02]**
(a) Model (b) Feature (c) Target (d)Training

Q.6 **Answer the following questions in details (Any Three).** **[12]**

1. Explain the difffernt layers of distributed data processing.

2. Explain the different architectures of distributed data processing.

3. Calculate the Degree Centrality of the following graph :



4. What is the significance of Singular Value Decomposition? Explain PCA technique of it with all steps and example.